Topic: Cybersecurity

Topic Title: Cybersecurity in Mobile Banking: Analyzing Phishing Attacks and Security Challenges in Bangladesh

Introduction

The rapid shift toward digital payments and mobile banking has transformed how people manage money globally (Boar & Szemere, 2020). In Bangladesh, this digital growth drives financial inclusion and strongly supports the country's "Smart Bangladesh" vision (Sarker & Khan, 2024). However, this rapid digital transformation has also significantly increased cybersecurity risks for both users and financial institutions (Waliullah et al., 2025). Phishing attacks, malware, and financial fraud frequently target commercial banks because they hold massive amounts of sensitive customer data and capital (Hossain et al., 2025).

To ensure the sustainable growth of digital banking, understanding these security challenges is now essential. This literature review analyzes the current state of mobile banking security by evaluating eight key studies. The review is organized into three major themes: the systemic vulnerabilities of digital banking in Bangladesh, the critical role of human behavior in phishing attacks, and the advanced technical frameworks proposed to mitigate these security risks.

Literature Review:

Sabikunnahar (24-59359-3)

Paper 1: SmartOTPs: An Air-Gapped 2-Factor Authentication for Smart-Contract Wallets

Overview: This paper introduces "SmartOTPs", an air-gapped two-factor authentication (2FA) framework specifically designed to secure smart-contract wallets without requiring an internet or network connection. To evaluate the practical efficiency of the system, the authors

analyze Ethereum gas costs and deployment trade-offs using a Merkle tree structure, demonstrating that caching strategies can significantly optimize rolling average costs per transaction. Furthermore, the paper compares SmartOTPs against existing hash-based approaches and popular hardware wallets like Trezor and Ledger, highlighting its usability, cost efficiency, and strong security resilience—even against potential post-quantum threats via Grover's algorithm. To prove its real-world viability, the researchers successfully implemented a prototype using Solidity smart contracts alongside a hardware authenticator built on a low-cost NodeMCU ESP8266 microcontroller.

Research Gap: Existing hardware wallets and 2FA mechanisms often suffer from vulnerability to online phishing attacks, high deployment costs, or heavy reliance on active network connectivity. This paper addresses these gaps by introducing a cost-effective, fully air-gapped authentication framework that eliminates internet dependencies while optimizing smart-contract gas fees through cryptographic trees.

Reference: Homoliak, I., Breitenbacher, D., Hujnak, O., Hartel, P., Binder, A., & Szalachowski, P. (2020). SmartOTPs: An air-gapped 2-factor authentication for smart-contract wallets. *Proceedings of the 2nd ACM Conference on Advances in Financial Technologies*, 21–36.

Tasfiqur Rahman (23-53951-3)

Paper 2: Cybercrime as a Threat to the Banking Sector: A Perspective from Commercial Banks in Bangladesh.

Overview: The banking sector in Bangladesh has become highly digital, making financial management easy but bringing new risks. Cybercrime is now a major threat to commercial banks because they hold massive funds and sensitive customer information. The authors report that cybercriminals exploit weaknesses in local banking systems using phishing, ransomware, malware, and DDoS attacks. These attacks threaten financial stability and expose system vulnerabilities. The study notes that many banks in Bangladesh still use legacy infrastructure, lack advanced security technologies, and invest very little in cybersecurity. To tackle these issues, the authors suggest implementing Security Information and Event Management (SIEM), Security Operation Centers (SOC), specialized security teams, and ongoing staff training.

Research Gap: There is a notable scarcity of studies on cybercrime and cybersecurity in Bangladesh's banking sector, especially from a comparative perspective. This paper examines cybercrime threats in both public and private banks but relies primarily on qualitative interviews with a limited sample size. The study lacks quantitative analysis regarding financial losses, attack frequency, and the direct effectiveness of implemented cybersecurity measures.

Reference: Hossain, H., Shohag, R. K., & Nath, N. C. (2025). Cybercrime as a threat to the banking sector: A perspective from commercial banks in Bangladesh. *Journal of Cybersecurity*, 12(1), 45–59.

Mostafa Wasef Arko (26-64633-1)

Paper 3: A New Taxonomy of Mobile Banking Threats, Attacks and User Vulnerabilities.

Overview: This paper focuses on mobile banking convenience and argues that the biggest weak spot is not the bank's servers or networks, but the user. The authors present a security framework split into four areas: the device, the network, physical data storage, and user behavior. They explain various attacks like phishing emails, fake login pages, man-in-the-middle attacks on Wi-Fi, and SQL injection. The real emphasis is on how bad user habits—like rooting/jailbreaking phones, using weak passcodes, clicking random links, or blindly accepting app permissions—create easy pathways for hackers. The paper recommends simple practices: do not trust urgent messages, do not root phones, check app permissions, and use strong passcodes. Securing mobile banking is a two-way responsibility.

Research Gap: The proposed taxonomy is conceptual and lacks empirical testing on real malware or live user data. There is no ranking of attacks by severity or impact, making defensive priorities unclear. Furthermore, no actual defense system is proposed (only recommendations), there is no user study to validate the behavioral claims, there is a limited comparison between Android vs. iOS vulnerabilities, and potential IoT/smart-city integration is mentioned but left unexplored. Additionally, the sources used are mostly pre-2018, missing newer mobile threat vectors.

Reference: Bojjagani, S., & Sastry, V. (2017). VAPTAI: A threat model for vulnerability assessment and penetration testing of Android and iOS mobile banking apps. In

Collaboration and Internet Computing (CIC), 2017 IEEE 3rd International Conference. [Additional reviews: C Insights (2015); Feizollah et al. (2015)].

Salman Farshi (25-60622-1)

Paper 4: A Review on E-Banking Security in Bangladesh: An Empirical Study.

Overview: This empirical study examines e-banking security in Bangladesh and identifies the major cyber threats facing the sector. It reviews major local security incidents, including ATM fraud, phishing, malware attacks, insider threats, and the Bangladesh Bank cyber heist. Key weaknesses identified include customer unawareness, outdated security infrastructure, inadequate automation, poor data privacy, and legacy systems. To address these, the authors propose a three-layer security framework consisting of process protection (multi-factor authentication), communication protection (secure VLANs), and server-end protection (intrusion prevention and hybrid cloud security). The study concludes that adopting modern security technologies and strengthening employee and customer awareness are critical.

Research Gap: Existing studies on e-banking security in Bangladesh focus mainly on individual security threats rather than looking at the entire banking ecosystem. There is a lack of comprehensive research that analyzes major cyber incidents, identifies infrastructure-wide vulnerabilities, and proposes an integrated security framework tailored specifically to Bangladesh's economic environment. Additionally, limited attention has been paid to insider threats and the active implementation of modern cybersecurity technologies.

Reference: Sijan, M. A. H., Shahoriar, A., Salimullah, M., Islam, A. S., & Khan, R. H. (2022). A review on e-banking security in Bangladesh: An empirical study. *In Proceedings of the 2nd International Conference on Computing Advancements (ICCA 2022)*. ACM.

Rafia Arefin Papry (23-52647-2)

Paper 5: Cybersecurity Considerations for Smart Bangladesh: Challenges and Solutions.

Overview: Cybersecurity is vital to protect computer networks and digital data from cyberattacks and unauthorized breaches. As Bangladesh advances toward its "Smart Bangladesh" vision, technologies like cloud computing, artificial intelligence (AI), IoT, e-governance, and online banking are expanding rapidly. While these technologies drive

economic growth, they also raise the risk of hacking, phishing, ransomware, and financial fraud. Weak security infrastructure, limited public awareness, and a lack of skilled cybersecurity professionals increase these risks. To secure the digital environment, this paper emphasizes the need for strong policies, legal frameworks, better technical infrastructure, and cooperation among the government, private sector, and academia.

Research Gap: While previous studies have discussed general cybersecurity threats, legal frameworks, and public awareness in Bangladesh, limited research has comprehensively examined the cybersecurity challenges and vulnerabilities specifically related to the unique infrastructure of the Smart Bangladesh initiative. This study addresses this gap by exploring the direct vulnerabilities and practical solutions needed to build a resilient, smart nation.

Reference: Sarker, M. A., & Khan, S. A. (2024). Cybersecurity considerations for Smart Bangladesh: Challenges and solutions. *Asian Journal of Research in Computer Science*, 17(6), 145–156.

Khaled Tanvir Shanto (26-64170-1)

Paper 6: Adoption of Digital Payments in Emerging Economies: Challenges and Policy Responses.

Overview: Digital payments have grown globally, moving trillions of dollars through mobile apps and e-wallets. This paper explores why this growth is vital and what challenges it faces. Digital payments are tools for economic growth and financial inclusion, yet cash remains popular, especially in poorer or rural areas where people hesitate to switch. Successful systems require real infrastructure: steady electricity, internet, reliable ID systems, and trusted agents. Users face major risks like unreliable networks, cash shortages at ATMs, confusing apps, and transaction fraud. These issues hurt user trust, which is the system's foundation. The authors suggest practical steps like monitoring agent liquidity, clearer user communication, stronger fraud protection, and using blockchain for safe transactions.

Research Gap: The paper reveals three major research gaps:

A methodological disconnect where behavioral research and macro-policy are siloed without standardized digital payment definitions.

A lack of real-world, empirical validation for proposed mitigation tools, blockchain scaling, and Central Bank Digital Currencies (CBDCs).

Isolated evidence that relies on localized cases rather than cross-country data, often ignoring the most vulnerable users and the systemic impact of Big Tech.

Reference: Boar, C., and Szemere, R. (2020). Payments go (even more) digital. *Bank for International Settlements*. [Additional references: CPMI Brief No 1 (2023); Ehrentraud et al. (2021)].

Hira Moni Mukta (25-61500-1)

Paper 7: Cyber Security Challenges and Mitigation Strategies in Digital Banking Industry of Bangladesh.

Overview: Digital banking, including internet banking and mobile financial services (MFS), has expanded rapidly in Bangladesh, boosting financial inclusion. However, this shift has also increased cybersecurity risks. Major challenges include phishing, ransomware, identity theft, and social engineering attacks. Financial institutions face these threats due to outdated software, weak authentication, and limited multi-factor authentication (MFA). There is also a shortage of skilled cybersecurity professionals and security weaknesses in mobile applications. The researchers recommend adopting AI-based threat detection, biometric authentication, MFA, regular updates, and public awareness programs to protect user information and secure the banking industry's sustainable growth.

Research Gap: Although previous studies identify major cybersecurity threats and suggest general mitigation strategies, there is very limited empirical research evaluating the real-world, practical effectiveness of these security measures. Additionally, more research is needed on customer awareness levels, the cybersecurity readiness of smaller financial institutions, and the hands-on implementation of advanced security technologies in Bangladesh.

Reference: Waliullah, M., George, M. Z. H., Hasan, M. T., Alam, M. K., Munira, M. S. K., & Siddiqui, N. A. (2025). Assessing the influence of cybersecurity threats and risks on the adoption and growth of digital banking: A systematic literature review. *arXiv*. [Additional reference: Mickey & Yanhaona (2024)].

Rakibul Rana Rolin (25-63473-3)

Paper 8: Modern Cybersecurity: Threats, Technologies, and Future Directions.

Overview: Cybersecurity is essential for protecting networks, devices, and sensitive information from data loss and unauthorized access. While technologies like Artificial Intelligence (AI), Machine Learning (ML), and blockchain are improving threat detection and defense systems, the increasing complexity of cyberattacks requires constant research. Systems across critical sectors (like banking, healthcare, and smart cities) face diverse threats including malware, ransomware, phishing, and cloud security risks. This study reviews current applications and challenges, emphasizing that future research must focus on developing smarter, more adaptive cybersecurity solutions to protect modern, interconnected digital systems.

Research Gap : Existing cybersecurity research has improved threat detection and protection through AI, ML, and blockchain technologies; however, gaps remain in defending against evolving cyber threats, securing IoT and cloud environments, and developing integrated, real-time, and scalable security frameworks. More research is required to create adaptive cybersecurity solutions that can effectively protect modern digital systems from emerging attacks.

Reference: Admass, W. S., Munaye, Y. Y., & Diro, A. A. (2023). Cyber security: State of the art, challenges and future directions. *Computers and Security: Artificial Intelligence*, vol. 1, Art. no. 100031.

Conclusion :

In conclusion, the literature shows that cybersecurity in mobile banking is an ongoing battle between rapid digital growth and evolving cybercrimes like phishing. The studies confirm that the biggest challenges in Bangladesh are outdated banking infrastructure, a shortage of local cybersecurity experts, and low user awareness. While advanced technologies like AI,

blockchain, and air-gapped authentication frameworks offer strong protection, they cannot succeed alone. Future security strategies must combine technical system upgrades with aggressive public awareness programs. Governments, financial institutions, and tech developers must collaborate to build a secure, dependable, and resilient mobile banking environment for the people of Bangladesh.